

STUDENT DATA ANALYSIS USING PANDAS

Internship Task 3 :

1. Title

Student Data Analysis using Pandas

2. Objective

The objective of this project is to analyze student data using Pandas.

The project demonstrates CSV loading, data cleaning, filtering, grouping, and generating summary statistics.

3. Problem Statement

Educational institutions generate large amounts of student data.

This project helps analyze student performance and generate useful insights using Python and Pandas.

4. Features Implemented

- CSV data loading
- Data cleaning
- Data filtering
- Grouping operations
- Summary statistics generation

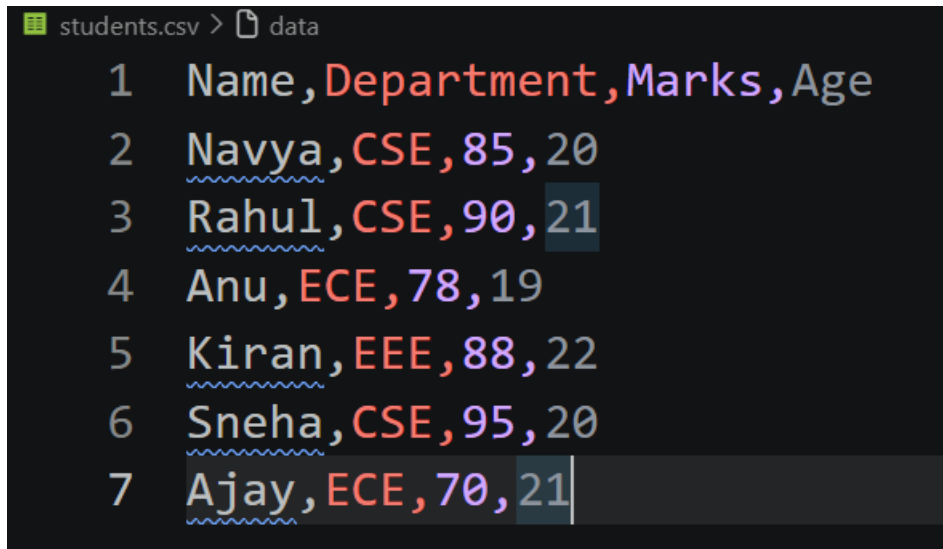
5. Technologies Used

Technology	Purpose
Python 3.x	Programming Language
Pandas	Data Analysis
CSV	Dataset Storage

6. Dataset Used

students.csv

Dataset contains: students Name, Department, Marks, Age



A screenshot of a code editor window with a dark background. The title bar shows 'students.csv' and a file icon. The editor displays the contents of the CSV file, with each line numbered from 1 to 7. The data is as follows:

	Name	Department	Marks	Age
1				
2	Navya	CSE	85	20
3	Rahul	CSE	90	21
4	Anu	ECE	78	19
5	Kiran	EEE	88	22
6	Sneha	CSE	95	20
7	Ajay	ECE	70	21

7. Source Code

```
1 import pandas as pd
2 df = pd.read_csv("students.csv")
3 print("\n----- STUDENT DATA -----")
4 print(df)
5 df = df.dropna()
6 average_marks = df["Marks"].mean()
7 print("\nAverage Marks:", average_marks)
8 highest_marks = df["Marks"].max()
9 print("Highest Marks:", highest_marks)
10 top_students = df[df["Marks"] > 80]
11 print("\n----- TOP STUDENTS -----")
12 print(top_students)
13 department_average = df.groupby("Department")["Marks"].mean()
14 print("\n----- DEPARTMENT AVERAGE -----")
15 print(department_average)
```

8. Sample Output

1. Install pandas : pip install pandas

```
(.venv) PS C:\Users\navya\Task3_Pandas> pip install pandas
Collecting pandas
  Downloading pandas-3.0.3-cp314-cp314-win_amd64.whl.metadata (19 kB)
Collecting numpy>=2.3.3 (from pandas)
  Downloading numpy-2.4.6-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.3-cp314-cp314-win_amd64.whl (9.9 MB)
  9.9/9.9 MB 1.3 MB/s 0:00:07
Downloading numpy-2.4.6-cp314-cp314-win_amd64.whl (12.5 MB)
  12.5/12.5 MB 1.7 MB/s 0:00:07
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, six, numpy, python-dateutil, pandas
Successfully installed numpy-2.4.6 pandas-3.0.3 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.2

[notice] A new release of pip is available: 26.0.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

2. Output:

```
Successfully installed numpy-2.4.6 pandas-3.0.3 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.2

[notice] A new release of pip is available: 26.0.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\navya\Task3_Pandas> python analysis.py

----- STUDENT DATA -----
   Name Department  Marks  Age
0  Navya         CSE     85   20
1  Rahul         CSE     90   21
2   Anu          ECE     78   19
3  Kiran          EEE     88   22
4  Sneha         CSE     95   20
5   Ajay          ECE     70   21

Average Marks: 84.33333333333333
Highest Marks: 95

----- TOP STUDENTS -----
   Name Department  Marks  Age
0  Navya         CSE     85   20
1  Rahul         CSE     90   21
3  Kiran          EEE     88   22
4  Sneha         CSE     95   20

----- DEPARTMENT AVERAGE -----
Department
CSE    90.0
ECE    74.0
EEE    88.0
Name: Marks, dtype: float64
(.venv) PS C:\Users\navya\Task3_Pandas>
```

9. Insights Generated

- Average student marks calculated
- Highest marks identified
- Top-performing students filtered
- Department-wise average generated

10. Conclusion

This project successfully demonstrates data analysis using Pandas.

The project includes CSV handling, cleaning, filtering, grouping, and generating meaningful insights from the dataset.